Will Buziak

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Education

Colorado School of Mines
M.S. Computer Science — *Graduation Date: December 2026*

University of Tennessee, Knoxville
B.S. Mechanical Engineering — *Graduation Date: May 2024*
Minor: Computer Science

Experience

Computer Architecture Design *Bahar Lab* — *Golden, CO*
Graduate Research Assistant — *Aug 2024 – Present*

- Secure Memory Architecture Design and Development for RISC-V Trusted Execution Environments.
- Modification and Development of the gem5 event-driven architecture simulator

A-4 Cybersecurity Department *Los Alamos National Lab* — *Los Alamos, NM*
Graduate Research Intern — *June 2026 – Aug 2026*

- AI Security research on Inference LLMs and hardware resource pattern analysis

Neuromorphic Computing *TENNLab* — *Knoxville, TN*
Undergraduate Research Assistant — *Dec 2022 – Aug 2024*

- Mathematical modeling for dual fuel diesel-ammonia combustion engines (collaborated with ORNL).
- Embedded computing and control applications using spiking neural networks.
- Robotic design for swarming robotics applications.

Electrochemical Energy Conversion and Storage Lab *EESCL* — *Knoxville, TN*
Undergraduate Research Assistant — *Aug 2022 – May 2024*

- Computational Fluid Dynamics (CFD) modeling of two-phase flow for hydrogen electrolyzer research.
- Web development for interactive dynamic data visualization tools.
- UI design for energy storage and power delivery system requirements.

Eck-Lectric Ind. — *Knoxville, TN*
Mechanical Engineering Intern — *Oct 2022 – Dec 2022*

- Assist in original product design for patent development.

Shaw Ind. — *Dalton, GA*
Process Engineering Co-op — *May 2022 – Aug 2022*

- Process improvement projects focused on waste management, safety, and process optimization.

Awards

- Graduate Living Expense Award Fall 2026
- C-MAPP Scholar 2025–2026
- EnergyTech University Prize 2023 Bonus Prize Finalist (Eck-Lectric)

IEEE / ACM Invited Talks

- YARCH '25 Workshop hosted by ASPLOS-EUROSYS 2025
- IEEE Denver Computer, Information Theory, and Robotics (CIR) - Spring 2025

Computer Skills

C/C++ | Python | Java | MATLAB | Solidworks/Onshape | Linux/Unix | Git | Raspberry Pi | Object-Oriented Programming | Shell Scripting | HTML/CSS | RISC-V | SLURM | SIMD | gem5 | QEMU | Qiskit | VHDL | Verilog | SystemVerilog | D-Wave | CUDA-Q | Event-Driven Simulators